$\cancel{1}$ $f(x,y) = e^{x^2-y^2}$

$$\frac{\partial f}{\partial x} = 2x \cdot e^{x^2-y^2}$$

$$\frac{\partial f}{\partial y} = -2y\, e^{x^2-y^2}$$

$$\nabla f = \begin{bmatrix} 2x \cdot e^{x^2-y^2} \\ -2y\, e^{x^2-y^2} \end{bmatrix}$$

$\cancel{2}$ $f(x,y) = x^2y - 3xy^2 + 7$

$$\frac{\partial f}{\partial x} = 2xy - 3y^2$$

$$\frac{\partial f}{\partial y} = x^2 - 6xy$$

$$\nabla f = \begin{bmatrix} 2xy - 3y^2 \\ x^2 - 6xy \end{bmatrix}_{(2,-1)} = \begin{bmatrix} -4 & -3 \\ 4 & +12 \end{bmatrix} = \begin{bmatrix} -7 \\ 16 \end{bmatrix}$$

3 $f(x,y) = x^2 - y^2$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = -2y$$

$$\boxed{\frac{\partial'' f}{\partial x''} = 2 \qquad \frac{\partial'' f}{\partial y''} = -2}$$

saddle point.

$\boxed{x=4}$ calc error

4 $D(x,y) = (x-1)^2 + (x-4)^2 + (x-7)^2$
$(y-2)^2 + (y-6)^2 + (y-2)^2$

$$\frac{\partial D}{\partial x} = 2(x-1) + 2(x-4) + 2(x-7)$$

$$\frac{\partial D}{\partial y} = 2(y-2) + 2(y-6) + 2(y-2)$$

$$\frac{\partial D}{\partial x} = 2(x-1) + 2(x-4) + 2(x-7)$$

$$\frac{\partial D}{\partial y} = 2(y-2) + 2(y-6) + 2(y-2)$$

$$\frac{\partial D}{\partial x} = 6x - 2 - 8 - 14 = 0$$

$$\Rightarrow 6x - 24 = 0$$

$$\boxed{x = 6}$$

$$\frac{\partial D}{\partial y} = 6y - 4 - 12 - 4 = 0$$

$$6y = 20$$

$$y = \frac{10}{3}$$

$$\boxed{\left(6, \frac{10}{3}\right)}$$

8 $f(x,y) = 5x^3y^2 - 4xy + 2y^4$

$$\frac{\partial f}{\partial x} = 15x^2y^2 - 4y$$

$$\frac{\partial f}{\partial y} = 10x^3y - 4x + 8y^3$$

6 $f(x,y) = x^3 - 3xy + y^3$

$$\frac{\partial f}{\partial x} = 3x^2 - 3y$$

$$\frac{\partial f}{\partial y} = -3y + 3y^2$$

critical points $\rightarrow$ $\frac{\partial f}{\partial y} = 0 \rightarrow \boxed{y = 0, 1}$

$$\boxed{x = 0, 1}$$

⑦ $T = 60 + 3(x-2)^2 + (y-4)^2$

$$\frac{\partial T}{\partial x} = 6(x-2)$$

$$\frac{\partial T}{\partial y} = 2(y-4)$$

$x = 2$

$y = 4$

$(2, 4)$

⑧ $f(x,y) = x^2 + y^2 - 4x + 6y + 8$

$$\frac{\partial f}{\partial x} = 2x - 4$$

$$\frac{\partial f}{\partial y} = 2y + 6$$

$$\frac{\partial f}{\partial x} = 2x - 4 \qquad \frac{\partial^2 f}{\partial x^2} = 2$$

$$\frac{\partial f}{\partial y} = 2y + 6 \qquad \frac{\partial^2 f}{\partial y^2} = 2$$

$$\boxed{\begin{aligned} x &= 2 \\ y &= -3 \end{aligned}}$$

$(2, -3) \rightarrow$ <u>minima</u>